Brief Report on Custom Artificial Neural Network (ANN) Architecture for Customer Churn Prediction

Why the Architecture Was Selected

The objective of this work was to develop two custom Artificial Neural Network (ANN) models for predicting customer churn while satisfying the given architectural requirements. Both models were designed with the same network structure so that the effect of different optimizers could be evaluated fairly. The architecture consisted of an input layer, three hidden layers, two Dropout layers, and an output layer.

The input layer contained **6 neurons**, corresponding to the processed input features after feature engineering and scaling. The **ReLU (Rectified Linear Unit)** activation function was selected because it is computationally efficient, helps overcome the vanishing gradient problem, and enables faster convergence during training. The **Glorot Uniform (Xavier Uniform)** weight initializer was used to maintain stable activation variance across layers, resulting in improved training performance.

The first hidden layer contained **64 neurons**, allowing the network to learn complex relationships and high-level patterns from the customer data. A **Dropout layer with a rate of 0.3** was added immediately after this layer to reduce overfitting by randomly deactivating 30% of the neurons during each training iteration. This encourages the model to learn more generalized features instead of memorizing the training data.

The second hidden layer consisted of **32 neurons** with the **Tanh activation function**. The reduction in neurons follows a hierarchical learning approach, where the network gradually compresses the learned information into more meaningful representations. The Tanh activation function outputs values between -1 and 1, which helps center the data around zero and can improve convergence in deeper networks. Another **Dropout layer with a rate of 0.2** was introduced after this layer to provide additional regularization and improve the model's ability to generalize to unseen data.

The third hidden layer contained **16 neurons** and again used the **ReLU activation function**. This layer further refined the extracted features before passing them to the output layer. The gradual reduction in neurons from 64 to 32 to 16 enabled the model to learn increasingly abstract representations while reducing computational complexity.

The output layer contained **one neuron** with the **Sigmoid activation function**, making it suitable for binary classification problems such as predicting whether a customer will churn or not. The Sigmoid function produces output values between 0 and 1, representing the probability of customer churn.

Both models were trained for **100 epochs** using a **batch size of 64**. These hyperparameters were selected because they provide a good balance between training stability, convergence speed, and overall model performance without causing excessive overfitting.

Impact of Each Modification

Although both ANN models shared the same architecture, they differed in the optimizer used during training. This allowed the study to evaluate the effect of optimization algorithms on the performance of the same neural network.

Model 1 – Adam Optimizer

The first model used the **Adam (Adaptive Moment Estimation)** optimizer. Adam automatically adjusts the learning rate for each parameter by combining the advantages of momentum and adaptive learning rate methods. This enables faster convergence, better optimization of the loss function, and reduced sensitivity to hyperparameter tuning.

The model achieved the following results:

- **Accuracy:** 84.65%
- **Precision:** 66.23%
- **Recall:** 49.38%
- **F1-Score:** 56.58%

The confusion matrix showed fewer misclassifications compared to the SGD model, indicating that Adam learned the underlying data patterns more effectively. The relatively higher recall demonstrated that the model successfully identified a larger proportion of customers likely to churn, which is particularly important for customer retention strategies.

Model 2 – SGD Optimizer

The second model used the **Stochastic Gradient Descent (SGD)** optimizer with default parameters. Unlike Adam, SGD updates model weights using gradient information from each mini-batch without automatically adapting the learning rate. As a result, it often requires careful tuning of learning rate and momentum to achieve optimal performance.

The model produced the following results:

- **Accuracy:** 81.30%
- **Precision:** 59.63%
- **Recall:** 23.70%
- **F1-Score:** 33.92%

Although SGD achieved reasonable precision, its recall was considerably lower than Adam. This indicates that many actual churn cases were incorrectly classified as non-churn, resulting in a higher number of false negatives. Consequently, the overall F1-score was also much lower, demonstrating weaker classification performance.

The inclusion of multiple hidden layers enabled both models to learn complex nonlinear relationships in the customer dataset. The combination of **ReLU and Tanh activation functions** improved feature learning, while the **Dropout layers** effectively reduced overfitting and enhanced generalization. However, the optimizer selection had the greatest influence on the final predictive performance.

Best Configuration and Why

Among the two custom ANN models, the configuration using the **Adam optimizer** produced the best overall performance.

The Adam-based model achieved the highest **accuracy (84.65%)**, indicating better overall prediction capability. More importantly, it achieved a significantly higher **recall (49.38%)** compared to SGD (23.70%), allowing it to correctly identify a much larger number of customers who were likely to churn. Since customer churn prediction focuses on identifying customers at risk of leaving, recall is one of the most critical evaluation metrics.

The Adam model also obtained the highest **F1-score (56.58%)**, demonstrating a better balance between precision and recall. This indicates that the model not only produced accurate predictions but also maintained consistent performance across different evaluation measures.

The superior performance of Adam can be attributed to its adaptive learning rate mechanism, which allows each network parameter to be updated efficiently during training. This leads to faster convergence, improved optimization, and better generalization without requiring extensive hyperparameter tuning. In contrast, the default SGD optimizer converged more slowly and struggled to capture important churn patterns, resulting in lower recall and overall performance.

Overall, both models successfully satisfied the required ANN architecture, including three hidden layers, multiple activation functions, and Dropout regularization. However, the experimental results clearly demonstrate that the **Adam optimizer is the better configuration** for this customer churn prediction problem because it provides higher accuracy, better recall, a stronger F1-score, and more reliable overall classification performance.